SOFTWARE REQUIREMENTS SPECIFICATION

Smart Hospital Management System

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Document Control

Field	Details
Document title	Software Requirements Specification (SRS) — Smart Hospital Management System
Version	1.0
Status	Draft — requirements should be validated with the project supervisor and intended users.
Intended audience	Project team, faculty/supervisor, developers, testers, and representative hospital stakeholders.
Standards note	Organized using a conventional IEEE-style SRS structure inspired by IEEE 830 and ISO/IEC/IEEE 29148. This is an academic template, not a claim of formal standards certification.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the proposed Smart Hospital Management System (SHMS). It explains what the system is expected to do, who will use it, and the main requirements that should guide its design and development. The document is prepared for an academic Software Engineering project and should be reviewed with the project supervisor before implementation.

1.2 Scope

SHMS is a web-based or mobile-accessible application for patients and authorized hospital personnel. It centralizes administrative and selected clinical workflows. Patients can manage profiles, request appointments, view permitted records, receive reminders, and request teleconsultations where enabled. Doctors can manage schedules, review patient history, document consultations, and create prescriptions. Nurses can record vital signs. Administrators manage users, departments, and system reports. An AI triage component may summarize reported symptoms, suggest a care specialty, and flag potentially urgent inputs for human review. It is decision support only and must not independently diagnose, prescribe, or replace emergency services or professional clinical judgment.

1.3 Intended Audience

- Project supervisor and academic evaluators.
- Software developers and UI/UX designers.
- Quality assurance and testing team.
- Hospital administrators, doctors, nurses, and patient representatives.
- Security, privacy, and deployment personnel.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SHMS	Smart Hospital Management System
SRS	Software Requirements Specification
AI	Artificial Intelligence
RBAC	Role-Based Access Control
MFA	Multi-Factor Authentication
PHI	Protected/Personal Health Information
Audit log	Time-stamped record of security-relevant or important system activity
Triage	Initial prioritization or routing based on reported symptoms; not a definitive diagnosis
Teleconsultation	Remote consultation between a patient and a qualified healthcare professional

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications (legacy structure reference).
- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
- Applicable local healthcare, privacy, electronic-record, accessibility, and cybersecurity requirements, as determined for the deployment jurisdiction.

1.6 Document Overview

Section 2 describes the product context and users. Section 3 lists functional requirements. Sections 4–6 define interfaces, quality requirements, and data. Sections 7–11 cover constraints, use cases, verification, traceability, and open issues.

2. Overall Description

2.1 Product Perspective

SHMS is a centralized application with a client interface, application/API layer, persistent database, and optional external services for messaging, video consultation, and AI-based triage. It may be deployed on a hospital-managed server or a compliant cloud environment. Integrations are configuration-dependent and are not assumed to exist until approved and implemented.

2.2 Product Functions

- Register and authenticate users and assign role-based permissions.
- Maintain patient profiles and authorized medical histories.
- Search doctors and departments, view available slots, and create, reschedule, or cancel appointments.
- Support doctor consultation notes, diagnoses, and prescriptions.
- Allow nurses or authorized clinicians to record vital signs.
- Provide AI-assisted symptom triage and specialty suggestions with clear limitations and escalation messaging.
- Send appointment confirmations, reminders, and relevant alerts.
- Generate bills, record payment status, and provide receipts where payment processing is enabled.
- Allow administrators to manage users, departments, schedules, and operational reports.
- Record audit events for sensitive access and important data changes.

2.3 User Classes and Characteristics

User class	Typical permissions and needs
Patient	Manage own profile; request appointments; view own permitted records, prescriptions, bills, and reminders.
Doctor	View assigned appointments and relevant patient history; record clinical notes and diagnoses; issue prescriptions.
Nurse	Access assigned patients as authorized; record vital signs and nursing observations.
Hospital administrator	Manage users, roles, departments, schedules, and operational reports; access clinical data only where explicitly authorized.
Receptionist / front-desk staff	Register patients, schedule appointments, and update administrative details without unrestricted clinical access.
System operator	Maintain deployment, backups, monitoring, and technical configuration; clinical access is not granted by default.

2.4 Operating Environment

- Modern browsers on desktop and mobile devices; a dedicated mobile app may be added if required.
- Server-side application/API and a relational or document-oriented database.
- HTTPS/TLS for network communication.
- Optional email/SMS/push notification provider and video consultation provider.
- Optional AI service that can be disabled if unavailable or not approved.

2.5 Design and Implementation Constraints

- Access to patient information must follow least-privilege and role-based authorization.
- AI-generated triage output must be presented as informational decision support and reviewed by qualified professionals where clinically relevant.
- The system must not expose passwords in plaintext; credentials must be securely hashed using an appropriate password-hashing algorithm.
- The deployment must meet applicable privacy, retention, and healthcare regulations for its jurisdiction.
- Availability of third-party services depends on their contracts, APIs, and connectivity.

2.6 User Documentation

The project should provide a short patient guide, staff/admin guide, deployment/configuration guide, and troubleshooting instructions.

2.7 Assumptions and Dependencies

- Users have access to a supported device and internet or hospital network connectivity.
- Hospital management supplies valid department, doctor, schedule, and billing data.
- The hospital defines which records each role may access and the applicable retention period.
- External messaging, payment, teleconsultation, or AI integrations are optional and depend on credentials and service availability.

3. System Features and Functional Requirements

Requirement IDs are unique and shall be used for design, implementation, and test traceability. Unless stated otherwise, requirements use “shall” to indicate mandatory behavior.

3.1 User Authentication and Authorization

ID	Requirement
FR-001	The system shall allow a user to register or be registered with the minimum required identity and contact details.
FR-002	The system shall authenticate users before granting access to protected functions.
FR-003	The system shall enforce role-based permissions for patients, doctors, nurses, administrators, and front-desk staff.
FR-004	The system shall allow users to log out and shall expire inactive sessions according to configurable security settings.
FR-005	The system shall provide a secure account-recovery process without revealing whether an arbitrary account exists.
FR-006	The system shall log security-relevant events, including repeated failed sign-in attempts and role or permission changes.

3.2 Patient Profile and Medical History

ID	Requirement
FR-007	The system shall allow an authorized patient or staff member to create and update a patient profile.
FR-008	The system shall store patient identifiers, contact details, date of birth, blood group if supplied, allergies if supplied, and relevant clinical history.
FR-009	The system shall allow a patient to view only their own records, except where an explicitly authorized proxy or caregiver arrangement exists.
FR-010	The system shall allow authorized clinicians to view relevant patient history for care purposes.
FR-011	The system shall retain a timestamp and author identifier for clinical record entries and updates.

3.3 Appointment and Scheduling Management

ID	Requirement
FR-012	The system shall display departments, doctor specializations, and available appointment slots.
FR-013	The system shall allow a patient or authorized staff member to request an appointment by selecting a doctor or specialty, date, and available time slot.
FR-014	The system shall prevent two confirmed appointments from reserving the same doctor slot unless the schedule explicitly supports a defined capacity greater than one.
FR-015	The system shall assign a unique appointment identifier and store appointment status, visit type, patient, doctor, and scheduled time.
FR-016	The system shall allow authorized users to reschedule or cancel appointments according to configurable policies.
FR-017	The system shall notify relevant users when an appointment is created, changed, or cancelled.
FR-018	The system shall allow doctors and authorized staff to view their appointment schedules.

3.4 AI-Assisted Symptom Triage

ID	Requirement
FR-019	The system shall allow a patient to submit symptoms and related information for optional AI-assisted triage.
FR-020	The triage service shall return a suggested specialty or next-step guidance with a clear statement that it is not a diagnosis.
FR-021	The system shall display urgent-care guidance when configured safety rules identify potentially emergency symptoms.
FR-022	The system shall not allow AI triage to issue prescriptions, confirm a diagnosis, or autonomously alter a treatment plan.
FR-023	The system shall identify when the triage service is unavailable and offer a non-AI path to appointment booking or contacting hospital staff.
FR-024	The system shall avoid presenting triage results as certain and shall provide an escalation path to a qualified clinician.

3.5 Consultation, Medical Records, and Prescriptions

ID	Requirement
FR-025	The system shall allow an authorized doctor to record consultation notes, diagnosis, and a treatment plan.
FR-026	The system shall allow an authorized doctor to create a prescription containing medicine name, dosage, frequency, duration, and instructions where applicable.
FR-027	The system shall associate clinical entries and prescriptions with the relevant patient and consultation or appointment.
FR-028	The system shall prevent unauthorized users from creating or modifying clinical records and prescriptions.
FR-029	The system shall maintain an audit history of clinical record creation and changes.

3.6 Vital-Sign Monitoring

ID	Requirement
FR-030	The system shall allow authorized nurses or clinicians to record vital signs, including configured measures such as temperature, pulse, blood pressure, and oxygen saturation.
FR-031	Each vital-sign entry shall include patient, measurement value and unit, timestamp, and recording user or device where available.
FR-032	The system shall display historical vital-sign readings to authorized clinicians.
FR-033	If configured thresholds are crossed, the system shall flag the reading for review and shall not represent an automated flag as a definitive diagnosis.

3.7 Notifications and Reminders

ID	Requirement
FR-034	The system shall send appointment confirmation after successful booking.
FR-035	The system shall support configurable reminders before an appointment through enabled channels.
FR-036	The system shall notify relevant users of schedule changes or cancellations.
FR-037	The system shall record notification status where the delivery provider exposes it and shall handle provider failures gracefully.

3.8 Billing and Payments

ID	Requirement
FR-038	The system shall allow authorized staff to generate an itemized bill associated with a patient and, where relevant, an appointment or service.
FR-039	The system shall track payment status, amount, issue date, and transaction reference when available.
FR-040	If an online payment provider is integrated, the system shall verify payment status through a trusted provider response rather than relying only on a client-side success message.
FR-041	The system shall provide a receipt or payment summary after a payment is recorded.

3.9 Administration and Reporting

ID	Requirement
FR-042	The system shall allow authorized administrators to create, update, deactivate, and assign roles to user accounts.
FR-043	The system shall allow authorized administrators to manage departments, doctor profiles, and doctor availability.
FR-044	The system shall provide operational reports such as appointment counts, cancellations, and billing summaries based on the user's permissions.
FR-045	The system shall log administrative changes to roles, departments, schedules, and configuration.

3.10 Teleconsultation (Optional Feature)

ID	Requirement
FR-046	If teleconsultation is enabled, the system shall allow an eligible patient to join a scheduled remote consultation using an approved provider.
FR-047	The system shall show joining instructions and the appointment time to authorized participants.
FR-048	The system shall not record audio or video by default; any recording feature must require explicit policy, consent, and appropriate access controls.

4. External Interface Requirements

4.1 User Interface

- The interface shall provide role-specific dashboards and clear navigation.
- Forms shall label required fields, validate input, and explain errors in understandable language.
- The interface shall provide confirmation before consequential actions such as cancellation or deletion where deletion is permitted.
- The interface shall be responsive for common desktop, tablet, and mobile screen sizes.
- AI-generated guidance shall be visually distinguishable from clinician-authored advice.

4.2 Hardware Interfaces

The baseline system does not require direct hardware integration. If medical devices or IoT sensors are integrated, the supported device models, communication protocol, calibration, units, timestamps, validation, and failure behavior must be specified and tested separately.

4.3 Software Interfaces

Interface	Purpose / requirement
Database	Persist users, appointments, clinical records, prescriptions, vital signs, bills, and audit events.
Email/SMS/push provider	Deliver confirmations, reminders, and alerts when configured.
AI triage service	Accept minimum necessary symptom input and return structured, safety-bounded guidance.
Payment gateway (optional)	Process payment securely and return verifiable transaction status.
Video consultation provider (optional)	Provide remote meeting links or session access.
Identity provider (optional)	Support institutional single sign-on if configured.

4.4 Communications Interfaces

Client-server traffic shall use HTTPS/TLS. External API calls shall use authenticated, documented interfaces. Secrets and API keys shall not be embedded in client-side code or source repositories.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-001	Under the agreed normal test load, 95% of routine page or API responses shall complete within 3 seconds, excluding third-party processing time.
NFR-002	The system shall support at least 100 concurrent authenticated users in the initial academic deployment; the target shall be revisited for production use.
NFR-003	Appointment booking shall prevent duplicate slot allocation under concurrent requests.

5.2 Security and Privacy

ID	Requirement
NFR-004	The system shall enforce least-privilege access and validate authorization on the server for every protected operation.
NFR-005	Passwords shall be stored only as secure salted hashes; plaintext passwords shall never be logged.
NFR-006	Sensitive data shall be transmitted over encrypted channels and protected at rest where supported by the deployment.
NFR-007	The system shall maintain tamper-resistant or access-controlled audit logs for sensitive access and changes.
NFR-008	The system shall minimize collection of personal and health data and restrict its use to stated purposes.
NFR-009	The system shall support configurable session expiry, account deactivation, backup access controls, and secret management.
NFR-010	The system shall provide a process for handling security incidents and suspected unauthorized disclosure.

5.3 Reliability and Availability

ID	Requirement
NFR-011	The system shall preserve committed data when a user refreshes or loses connection after a confirmed save.
NFR-012	The initial deployment target shall be 99.5% monthly availability, excluding planned maintenance and external-provider outages, if hosted in an environment capable of meeting this target.
NFR-013	If an optional external service fails, core patient registration and appointment workflows shall remain usable where technically feasible.
NFR-014	The project shall define and test backup and restoration procedures before deployment.

5.4 Usability and Accessibility

ID	Requirement
NFR-015	A first-time user shall be able to complete basic registration and appointment booking with clear on-screen guidance.
NFR-016	The interface shall use consistent labels, readable typography, and understandable error messages.
NFR-017	The interface should target WCAG 2.1 AA accessibility practices, including keyboard navigation, labels, contrast, and text alternatives.

5.5 Maintainability and Portability

ID	Requirement
NFR-018	The application shall separate presentation, business logic, and data-access responsibilities sufficiently to support testing and maintenance.
NFR-019	The system shall provide documented configuration and deployment steps, including environment variables and database setup.
NFR-020	The system shall record application errors without unnecessarily exposing sensitive health information in logs.

5.6 AI Safety and Transparency

ID	Requirement
NFR-021	The AI triage interface shall clearly state its intended use, limitations, and non-diagnostic status.
NFR-022	The system shall provide a human escalation route and a safe fallback when the AI service fails or returns an unusable result.
NFR-023	The project shall document the intended inputs, output categories, known limitations, and evaluation approach for the AI feature before release.

6. Data Requirements

6.1 Core Data Entities

Entity	Key data fields (illustrative)
User	userId, name, contact details, credential hash, role, account status, createdAt
Patient	patientId, userId, dateOfBirth, bloodGroup (optional), allergies (optional), emergency contact (if required)
Doctor	doctorId, userId, specialization, license number, departmentId, availability
Department	departmentId, name, location, contact details
Appointment	appointmentId, patientId, doctorId, dateTime, visitType, status, reason (optional), createdAt
MedicalRecord	recordId, patientId, authorId, appointmentId (optional), diagnosis, notes, createdAt, updatedAt
Prescription	prescriptionId, patientId, doctorId, recordId, medication, dosage, frequency, duration, instructions
VitalSigns	vitalId, patientId, recordedBy, measurement type, value, unit, timestamp
TriageResult	resultId, patientId, input reference or minimized summary, suggested specialty, urgency level, generatedAt, model/version
Bill / Payment	billId, patientId, appointmentId (optional), amount, paymentStatus, issuedAt, transaction reference (optional)
Notification	notificationId, recipientId, channel, message template/reference, status, sentAt
AuditEvent	eventId, actorId, action, resource type/id, timestamp, outcome

6.2 Data Integrity

- Every entity shall have a stable unique identifier.
- References to patients, doctors, appointments, and records shall be validated before a transaction is committed.
- Appointment slot reservation and confirmation shall be handled atomically or with equivalent concurrency protection.
- Clinical record changes shall preserve author and timestamp metadata.
- Units and timestamps shall be stored consistently; the display timezone shall be defined for the hospital.

6.3 Data Retention and Deletion

Retention periods, patient access/correction rights, deletion rules, consent requirements, and legal holds must be defined by the hospital and applicable law. Clinical records must not be deleted merely because a user account is deactivated when retention obligations apply.

7. Business Rules and Constraints

- BR-001: A patient may access only their own health data unless an explicitly authorized legal representative or caregiver permission applies.
- BR-002: Only qualified and authorized clinicians may enter or amend diagnoses and prescriptions.
- BR-003: An appointment is confirmed only after the selected slot is successfully reserved and persisted.
- BR-004: Cancellation and rescheduling policies shall be configurable by the hospital.

- BR-005: AI triage output is advisory; emergency guidance must not delay contacting local emergency services.
- BR-006: Administrative privileges shall not automatically grant unrestricted clinical access.
- BR-007: Any online payment integration shall rely on trusted server-to-server verification of payment status.
- BR-008: Access to health information and significant changes shall be auditable.

8. Use Cases

8.1 UC-01: Book an Appointment

Field	Description
Primary actor	Patient; receptionist may act on behalf of a patient under hospital policy.
Preconditions	User is authenticated or the hospital permits assisted booking; doctor schedules are available.
Trigger	Patient selects “Book Appointment”.
Main flow	1. System displays departments/doctors and slots. 2. Patient selects a specialty or doctor and slot. 3. System validates eligibility and slot availability. 4. System reserves the slot and creates an appointment. 5. System displays a confirmation and queues notifications.
Alternative flow	If the slot has been taken, the system informs the patient and displays current alternatives. If notification delivery fails, booking remains saved and the failure is logged.
Postconditions	A unique appointment exists with confirmed status, or no appointment is created.

8.2 UC-02: AI-Assisted Symptom Triage

Field	Description
Primary actor	Patient.
Preconditions	Triage feature is enabled and its safety notice is available.
Trigger	Patient chooses symptom guidance.
Main flow	1. System explains limitations. 2. Patient enters symptoms. 3. Service evaluates input. 4. System presents a suggested specialty/next step and non-diagnostic disclaimer. 5. Patient may proceed to booking or contact hospital staff.
Alternative flow	If emergency indicators are identified, show urgent-care guidance. If AI is unavailable, show a fallback path. Do not present a failed or incomplete response as a valid result.
Postconditions	A triage result may be stored according to policy and can be linked to a subsequent appointment with appropriate access controls.

8.3 UC-03: Record Consultation and Prescription

Field	Description
Primary actor	Doctor.
Preconditions	Doctor is authenticated and authorized; patient identity is verified.
Trigger	Doctor opens an assigned appointment or patient record.
Main flow	1. System displays permitted relevant history. 2. Doctor records consultation notes and diagnosis. 3. Doctor creates a prescription if clinically indicated. 4. System validates required fields and saves entries with author and timestamps. 5. Patient is granted access to the permitted information.
Alternative flow	If save fails, system informs the doctor and does not show a false success state. Unauthorized access is denied and logged.
Postconditions	Consultation data is saved and linked to the patient; prescription exists only if created by an authorized doctor.

9. Acceptance Criteria and Verification

Requirement area	Verification / acceptance criterion
Authentication and roles	Tests confirm each role can access permitted functions and is denied access to unauthorized functions.
Appointment booking	A valid booking receives a unique ID; two concurrent attempts for one exclusive slot cannot

	both succeed.
Medical records	Authorized clinician can create a record; unauthorized patient or staff account cannot view or alter another patient's record.
AI triage	UI displays non-diagnostic notice, provides fallback on service failure, and displays configured urgent-care guidance for test cases.
Vital signs	Recorded values include units, timestamp, patient, and recorder; configured out-of-range values are flagged for review.
Notifications	Booking and cancellation events enqueue the appropriate notification; provider failure is handled without losing the appointment.
Billing	Bill totals and payment status are consistent; online payment status is verified server-side if integrated.
Security and audit	Sensitive actions create audit events; passwords and unnecessary health details are not written to application logs.
Usability	Representative users complete core tasks using the interface without blocking usability defects.
Backup and recovery	A documented backup can be restored in a test environment and integrity checks pass.

Verification methods may include inspection, analysis, demonstration, unit testing, integration testing, system testing, security testing, and user acceptance testing.

10. Traceability Matrix

Feature	Requirement IDs	Representative verification
Authentication / RBAC	FR-001–FR-006; NFR-004–NFR-010	Role/permission and security tests
Patient profile / records	FR-007–FR-011; FR-025–FR-029	CRUD, access-control, and audit tests
Appointment scheduling	FR-012–FR-018; NFR-003	Booking, cancellation, concurrency tests
AI triage	FR-019–FR-024; NFR-021–NFR-023	Scenario tests, failure fallback, safety review
Vital-sign monitoring	FR-030–FR-033	Validation, units, threshold tests
Notifications	FR-034–FR-037	Provider integration and failure tests
Billing	FR-038–FR-041	Calculation and payment reconciliation tests
Administration	FR-042–FR-045	Role and audit tests
Teleconsultation	FR-046–FR-048	Integration and consent/access tests
Quality / operations	NFR-001–NFR-020	Performance, usability, resilience, backup and security tests

11. Assumptions, Dependencies, and Open Issues

11.1 Assumptions

- This SRS describes a proposed academic system; specific hospital policies have not yet been supplied.
- AI triage, online payment, teleconsultation, and direct device integration may be optional modules.
- Initial performance targets are provisional and must be confirmed for the expected deployment.

11.2 Dependencies

- Hosting infrastructure, database, domain, TLS certificates, and backup storage.
- Any selected SMS/email/push, payment, video, or AI provider and its API availability.
- Hospital-provided staff, department, schedule, billing, and retention policies.

11.3 Open Issues to Resolve Before Implementation

- Which country/state and healthcare privacy requirements govern the deployment?
- Will the application be web-only, mobile-only, or both?
- Which AI triage model/service will be used, and who approves its clinical safety?
- Which roles may view, export, correct, or download each category of patient data?
- Are payment and teleconsultation integrations required for the first release?

- What are the exact backup retention, recovery-time, and recovery-point targets?
- What patient consent and proxy/caregiver workflows are required?

Appendix A: Glossary

Term	Meaning
Authentication	Verifying the identity of a user or system.
Authorization	Determining what an authenticated user is permitted to do.
Audit trail	Chronological record of relevant actions and changes.
Functional requirement	A behavior or service the system must provide.
Non-functional requirement	A quality, constraint, or operational property the system must satisfy.
Traceability	Ability to link a requirement to design elements and verification evidence.

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